06 — HARDWARE · TEAM BUILD

ESP32 Walkie-Talkie

A working push-to-talk walkie-talkie, built from a bag of parts against the clock. Our team finished first.

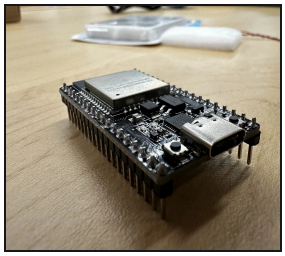


FIG. 6.1 — ESP32 development board

CONTEXT	ROLE	YEAR	METHODS
Workshop competition	Team member: circuit build, testing	2025	ESP32 · ESP-NOW · breadboarding · audio amp

1st	1	1
TEAM TO A WORKING MODEL	BREADBOARD, NO SOLDERING	BUTTON: HOLD TO TALK, RELEASE TO LISTEN

01 BRIEF

A timed workshop competition: turn a bag of parts into a walkie-talkie that actually works, before anyone else. Every team started with the same parts and the same clock. Nothing counted except a working model.

02 APPROACH

01 Build around the ESP32
The ESP32 pairs a fast processor with a 2.4 GHz radio, so one chip samples the audio, packs it into packets and transmits it.

02 Drive the speaker
A GPIO pin can't source the current a speaker needs, so received audio runs through a small amplifier board to a compact speaker.

03 Add push-to-talk
One button switches the unit between talking and listening: hold to talk, release to hear the other side.

The clock	FAILURE MODE 6.A
RISK	A hard time limit, with every team racing for the same finish line. A fix that took minutes could cost the win.
DECISION	No soldering. With the whole circuit on one breadboard, every correction was a wire moved and a re-test in seconds.
RESULT	First team to a working model.

04 REV. B

- Move to a soldered perfboard with a battery, so it works away from a laptop.
- Design a handheld enclosure in Fusion 360 and 3D-print it.

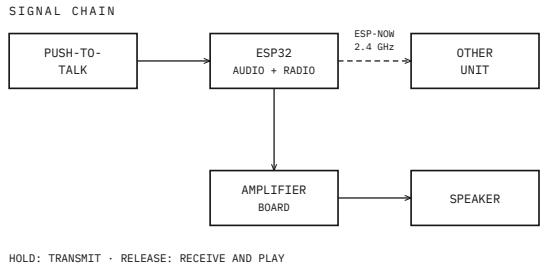


FIG. 6.2 — Signal chain of one unit.

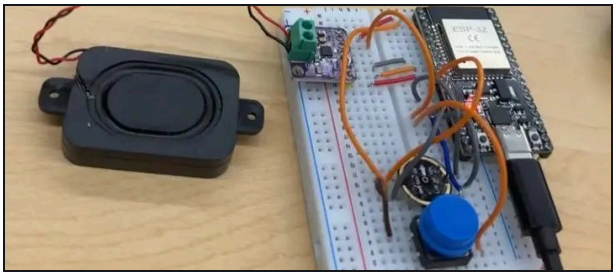


FIG. 6.3 — The finished prototype: ESP32, amplifier, speaker and push-to-talk button on one breadboard. A video of it working is on the portfolio site.

03 HOW THE LINK WORKS

Builds like ours talk over **ESP-NOW**, Espressif's connectionless peer-to-peer protocol. It uses the ESP32's Wi-Fi radio in the 2.4 GHz band, the same band as Bluetooth, but needs no router, pairing or network: the talking unit broadcasts short packets of audio samples (up to 250 bytes each) and every unit on the same channel receives them, much like a handheld radio shares a frequency.

Near-zero setup and low latency suit a race against the clock. The trade-off is unacknowledged delivery: a dropped packet is a brief click, not a resend.

> *Under a deadline, the design that's fastest to change beats the one that's neatest on paper.*